

# Inline and Display Mathematics

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$\gamma_{g,9} : |D_{\text{inv}}\rangle \rightarrow \frac{1}{\sqrt{D}} \sum_{\alpha} \sum_{\beta, \gamma} c_{\alpha\beta} \bar{c}_{\gamma\alpha} |d_{\beta} \otimes \bar{d}_{\gamma}\rangle = \frac{1}{\sqrt{D}} \sum_{\alpha} |d_{\alpha} \otimes \bar{d}_{\alpha}\rangle = |D_{\text{inv}}\rangle ,$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$\lim_{r \rightarrow \infty} \int_{\mathcal{I}^-} dt d\phi dt' d\phi' \left[ \frac{(rr')^2}{l^2} \left( \Phi(r, t, \phi) \overset{\leftrightarrow}{\partial}_{r_*} G(r, t, \phi; r', t', \phi') \overset{\leftrightarrow}{\partial}_{r'_*} \Phi(r', t', \phi') \right) \right]_{r=r'} ,$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$\langle E_{eff}(x_0, x_1) \rangle = \frac{1}{6} \left[ \frac{\partial^2}{\partial x_0^2} \frac{\partial^2}{\partial y_1^2} - \frac{\partial^2}{\partial x_0 \partial x_1} \frac{\partial^2}{\partial y_0 \partial y_1} \right] \Delta_{\varphi}(x_0, x_1; y_0, y_1)|_{y_{\mu} \rightarrow x_{\mu}} ,$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$\frac{\delta}{\delta\phi_A^*} + \frac{i}{\hbar} \frac{\delta\Gamma}{\delta\phi_A^*} - (-1)^{\varepsilon_i(\varepsilon_A+1)} (\Gamma''-1)^{ij} \left( \frac{\delta_l}{\delta\varphi^j} \frac{\delta\Gamma}{\delta\phi_A^*} \right) \frac{\delta_l}{\delta\varphi^i} ,$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$\hat{h}_{\text{red}} = \sum_{\vec{b}} \left\{ -\frac{g^2 LM}{4\sqrt{2}} \frac{1}{J} \frac{\delta}{\delta\varphi_{\vec{b}}} J \frac{\delta}{\delta\varphi_{\vec{b}}} + \left( \frac{4\sqrt{2}}{g^2 LM} \right) \sum_{\vec{\varepsilon}} (\varphi_{\vec{b}} - \varphi_{\vec{b}+\vec{\varepsilon}})^2 \right\} .$$